

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12397522
B2
August 26, 2025
Bardin; Damien

Curing mold for a tire, comprising means for cutting a rubber mold seam

Abstract

A curing mold (**10**) for a tire comprises: a plurality of circumferentially distributed sectors (**18**) that are intended to mold the tread of the tire, at least one sipe blade (**40**) intended to mold a void in the tread, the sipe blade extending over two adjacent sectors (**18**) such that the sipe blade comprises a part (**40a**, **40b**) in each of the two sectors, the two parts of the sipe blade being separated, in the closed position of the mold, by a space (**26**) that creates flash during the curing of a tire, the parts of the sipe blade comprising cutting means (**38**) that are able to cut off the flash during the opening of the mold.

Inventors:	Bardin; Damien (Clermont-Ferrand, FR)
Applicant:	COMPAGNIE GENERALE DES ETABLISSEMENTS MICHELIN (Clermont-Ferrand, FR)
Family ID:	1000008777830
Assignee:	COMPAGNIE GENERALE DES ETABLISSEMENTS MICHELIN (Clermont-Ferrand, FR)
Appl. No.:	18/021853
Filed (or PCT Filed):	August 16, 2021
PCT No.:	PCT/FR2021/051468
PCT Pub. No.:	WO2022/038328
PCT Pub. Date:	February 24, 2022

Prior Publication Data

Document Identifier

US 20230356489 A1

Publication Date

Nov. 09, 2023

Foreign Application Priority Data

FR

2008560

Aug. 19, 2020

Publication Classification

Int. Cl.: **B29D30/06** (20060101); **B29C37/00** (20060101); **B29D30/68** (20060101); **B29L30/00** (20060101)

U.S. Cl.:

CPC **B29D30/0606** (20130101); **B29C37/00** (20130101); **B29D30/0681** (20130101); **B29D30/68** (20130101); **B29C2793/0054** (20130101); **B29D2030/0613** (20130101); **B29L2030/006** (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3666852	12/1971	Burke	29/527.1	B29C 70/84
6318983	12/2000	Lopez et al.	N/A	N/A
9415555	12/2015	Ohashi	N/A	N/A
10525532	12/2019	Desvignes et al.	N/A	N/A
11358357	12/2021	Gueugneau et al.	N/A	N/A
2007/0284783	12/2006	Ohara	N/A	N/A
2015/0165706	12/2014	Gaebelein	264/293	B29D 30/0629
2016/0129651	12/2015	Ohashi	N/A	N/A
2017/0216922	12/2016	Desvignes et al.	N/A	N/A
2018/0065332	12/2017	Kaneko	N/A	N/A
2019/0184659	12/2018	Gueugneau et al.	N/A	N/A
2023/0166468	12/2022	Bardin et al.	N/A	N/A
2023/0347608	12/2022	Bardin et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
19506697	12/1995	DE	N/A

102018213083	12/2019	DE	N/A
579436	12/1993	EP	B29D 30/0606
2813354	12/2013	EP	N/A
3263307	12/2017	EP	N/A
3024059	12/2015	FR	N/A
3-279006	12/1990	JP	N/A
10-264271	12/1997	JP	N/A
2019181848	12/2018	JP	N/A
20-0408355	12/2005	KR	N/A
2017/221283	12/2016	WO	N/A

OTHER PUBLICATIONS

Yonezu. JP2019181848A. 2019. ESpaceNet Machine Translation. (Year: 2025). cited by examiner
International Search Report dated Nov. 18, 2021, in corresponding PCT/FR2021/051468 (6 pages).
cited by applicant

Primary Examiner: Hauth; Galen H

Assistant Examiner: Booth; Alexander D

Attorney, Agent or Firm: VENABLE LLP

Background/Summary

BACKGROUND

(1) The invention lies in the field of tyre manufacturing and relates more particularly to a mould for curing a tyre, having means for cutting off rubber flash.

(2) The manufacture of a tyre comprises a curing step during which a green tyre is moulded and vulcanized so as to give the tyre its characteristics and its final appearance. A tyre is cured in a curing mould. The mould comprises a plurality of moulding elements which form, in the closed position of the mould, an internal moulding surface corresponding to the negative of the external surface of the tyre. In particular, the mould comprises a plurality of sectors that are distributed circumferentially with respect to the axis of the mould and are intended to mould the tread of the tyre, and two shells that are disposed axially and are intended to mould the sidewalls of the tyre.

(3) During curing, a rubber-based green tyre is pressed against the internal moulding surface. At the same time, the moulding elements are kept in contact with one another by closing means in order to prevent the mould from opening under the effect of the pressure exerted by the green tyre against the internal moulding surface. Thus, during curing, the mould remains closed and the rubber under pressure does not creep between the moulding elements. However, it is possible for certain moulding elements to be separated by a very narrow space, resulting, for example, from the wearing of the contact surfaces between said moulding elements, or from a design choice relating to the need to evacuate the air trapped between the green tyre and the mould. The rubber under pressure is then able to creep into said space and to form, on the tyre, flash which may have a detrimental effect on its characteristics and on its final appearance.

(4) The application US2007284783A1 describes a mould in which each sector has, on its circumferential edges, a contact surface that comes to bear on the contact surface of the adjacent sector, and a surface slightly set back circumferentially and referred to as a “non-contact” surface,

which is disposed radially towards the inside of the mould and has a roughness of between 50 and 200 μm . During curing, the non-contact surfaces of two adjacent sectors move towards one another under the effect of their thermal expansion so as to come to bear against one another. Thus, the mould does not allow the rubber under pressure to creep between the sectors, while allowing the evacuation of the air by virtue of the roughness of the non-contact surfaces.

(5) However, the wearing of the contact surfaces between the sectors has the consequence of the non-contact surfaces getting closer together and the contact pressure between the non-contact surfaces increasing when they come into contact under the effect of the thermal expansion of the sectors, said pressure being able to bring about premature deterioration of the roughness of the non-contact surfaces by burnishing. By contrast, when the mould is no longer capable of compensating for the increasing closeness of the contact surfaces between the sectors on account of their wear, the formation of a space between the contact and non-contact surfaces is observed, said space being able to bring about the creation of flash.

(6) An objective of the invention is to overcome the drawbacks of the prior art and to provide an original solution in order to improve the characteristics and final appearance of the tyres, while avoiding premature deterioration of the moulding elements.

SUMMARY

(7) This objective has been achieved by the invention, which proposes, according to a first subject, a mould for curing a tyre, comprising at least two moulding elements, said moulding elements being separated, in the closed position of the mould, by a space that creates flash during the curing of a tyre, characterized in that said moulding elements comprise cutting means that are able to cut off said flash during the opening of the mould.

(8) In other words, it is preferable to maintain a very narrow space between certain moulding elements, even taking into account their thermal expansion, in order to avoid the deterioration of said moulding elements.

(9) In addition, during the opening of the mould, said moulding elements move away from the tyre along given trajectories. The cutting means of said moulding elements then cut off the flash created between the two moulding elements. The flash is cut off, torn off, or cracked so as to allow it to be eliminated during the running of the tyre, which then rapidly achieves characteristics and an appearance that are satisfactory.

(10) Advantageously, each of said moulding elements comprises an edge partially delimiting the space that separates them in the closed position, the cutting means being arranged on the edge of at least one of said moulding elements such that the cutting means cut off said flash during their withdrawal from the tyre, each of said moulding elements being able to be withdrawn during the opening of the mould. Such an arrangement of the cutting means is simple and makes it possible to cut off the flash effectively.

(11) Preferably, the cutting means are arranged on said edge of each of said moulding elements, the cutting means of each of said moulding elements having complementary shapes. Such shapes make it possible to improve the effectiveness of the cutting off of the flash.

(12) Advantageously, the cutting means are integral with the edge of at least one of said moulding elements. Such a design is simple and allows time to be saved during the manufacture of the moulding elements.

(13) Preferably, the cutting means have a recessed and/or protruding shape. Such a non-rectilinear shape makes it possible to extend the parts of the cutting means in contact with the flash and, consequently, to increase the effectiveness of the cutting off of the flash.

(14) FIG. 2 illustrates said space **36**, which, advantageously, has a length L of between 0.2 and 20 mm and, more preferably, between 0.5 and 5 mm. Preferably, said space **36** has a thickness E of between 0.1 and 3 mm, more preferably between 0.2 and 2 mm, and even more preferably between 0.3 and 1 mm.

(15) Such a space with restricted dimensions means having moulding elements that likewise have

restricted dimensions and consequently are fragile. Such dimensioning of the moulding elements makes the implementation of the invention particularly advantageous.

(16) Preferably, the mould comprises: a plurality of circumferentially distributed sectors that are intended to mould the tread of said tyre, at least one sipe blade intended to mould a void in said tread, said sipe blade extending over two adjacent sectors such that said sipe blade comprises a part in each of the two sectors, said moulding elements taking the shape of each part of the sipe blade. The sipe blade has a thickness of between 0.2 and 20 mm and, more preferably, between 0.5 and 5 mm. In the same way, such an arrangement and such dimensioning of the moulding elements make the implementation of the invention particularly advantageous.

(17) Preferably, the cutting means have the shape of a triangular or crenellated toothing. Such a shape makes it possible to improve the effectiveness of the cutting off of the flash.

(18) Advantageously, the cutting means are obtained by way of a process of consolidation by selective melting. The use of such a process, which is also employed in the manufacture of the moulding elements on which the cutting means are arranged, makes it possible to reduce the manufacturing cost and duration for the cutting means.

(19) A second subject of the invention is a method for manufacturing a tyre, comprising a step of curing a tyre in a curing mould according to the first subject of the invention.

(20) Lastly, a third subject of the invention is a tyre intended to be mounted on the rim of a wheel, the tyre being obtained by a manufacturing method according to the second subject of the invention.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The invention will be understood better from the rest of the description, which is based on the following figures:

(2) FIG. 1 is a schematic depiction of a curing mould according to the prior art;

(3) FIG. 2, described above, is a schematic depiction of certain elements of the mould in FIG. 1;

(4) FIG. 3 is a schematic depiction of certain elements of a curing mould according to a preferred embodiment of the invention.

DETAILED DESCRIPTION

(5) In the various figures, elements that are identical or similar bear the same reference signs. Their description is therefore not systematically repeated.

(6) FIG. 1 illustrates a schematic depiction of a curing mould **10** for moulding and curing a green tyre, in order to obtain a tyre with desired characteristics and a desired appearance. The mould delimits a cavity **11** that is generally rotationally symmetric about a central axis **12**. The mould comprises a frame, moulding elements, curing means, and means for opening and closing the mould. The opening and closing means are able to set the moulding elements in movement between an open position and a closed position. The open position corresponds to the step of withdrawing a moulded and vulcanized tyre, and to the step of introducing a new green tyre. The closed position (FIG. 1) corresponds to the step of moulding and curing the green tyre. In the closed position of the mould, the moulding elements form an internal moulding surface **14** corresponding to the negative of the desired external surface of the tyre. The curing means are able to provide the thermal energy and/or the pressure that are required, respectively, for the curing and/or the pressing of the green tyre against the internal moulding surface **14**. By way of example, thermal energy and pressure are provided by pressurized steam circulating about the mould, and inside the green tyre in an extendable membrane.

(7) Throughout the following text and unless indicated to the contrary, an axial direction denotes a direction parallel to the central axis **12**, a radial direction **16** denotes a direction perpendicular to

and intersecting the central axis **12**, and a circumferential direction denotes a direction perpendicular to a radial direction **16** and to the central axis **12**.

(8) The mould **10** more particularly comprises a plurality of sectors **18** intended to mould the tread of the tyre, two shells **20** intended to mould the sidewalls of the tyre, and two rings **22** intended to mould the beads of the tyre. In the closed position of the mould, the sectors **18** are distributed circumferentially with respect to the central axis **12** of the mould so as to form a ring, and are able to be withdrawn radially with respect to the tyre during the opening of the mould.

(9) FIG. 2 illustrates certain elements of the mould **10**, which comprises at least two adjacent moulding elements **24**, said moulding elements being separated, in the closed position of the mould, by a space **26** that creates flash (not shown) during the curing of a tyre. By way of example, the two moulding elements **26** may be two sectors **18**, a sector **18** and a shell **20**, or two elements that are used in the composition of one or more sectors, such as lining elements **28**, sipe blades **30** intended to mould grooves in the tread of the tyre, or sipe blades intended to mould voids in said tread. Each of said moulding elements **24** comprises a plurality of faces having the various functionalities. In particular, a moulding element comprises a moulding surface **32** contributing to the internal moulding surface **14** of the mould, a plurality of contact surfaces **34** or interfaces with adjacent moulding elements, and at least one edge **36** partially delimiting the space **26** that creates the flash. By way of example, the presence of said space **26** results from a design choice intended to avoid impacts between fragile parts of the two moulding elements **24**, said moulding elements coming into contact strongly via their contact surfaces **34** during the closure of the mould.

(10) FIG. 3 illustrates said moulding elements **24**, which comprise, according to the invention, cutting means **38** that are able to cut off said flash during the opening of the mould. During the opening of the mould, said moulding elements move away from the tyre along given trajectories. The cutting means of said moulding elements then cut off the flash created between the two moulding elements. The cutting means are arranged on the edge **36** of at least one of said moulding elements **24** such that said flash slides along the cutting means as they are withdrawn.

(11) The cutting means have a recessed and/or protruding shape. The recesses and the protrusions of the cutting means make it possible to lengthen the sliding path of the flash along the cutting means as they are withdrawn. By way of examples, the cutting means take the form of scissors, blades, spikes or any other suitable shape for cutting, tearing or forming an incipient crack in the flash.

(12) Preferably, the cutting means are arranged on the edge **36** of each of said moulding elements **24**, the cutting means of each of said moulding elements having complementary shapes. Thus, the effectiveness of the cutting off of the flash by the cutting means is improved. The cutting means **38** are integral with the edge **36** of at least one of said moulding elements **24**. By way of example, the cutting means may also be fixed to the edges **36** of the moulding elements **24** by any other suitable means.

(13) Said space **26** has a length L of between 0.2 and 20 mm and, more preferably, between 0.5 and 5 mm. The length of said space denotes, in a circumferential plane, the longest distance along which the two moulding elements are edge-to-edge. Thus, it is particularly appropriate to provide a space **26** and cutting means **38** on the edges of moulding elements **24**, the contact surface of which has a short extent and is, consequently, more fragile.

(14) Said space has a thickness E of between 0.1 and 3 mm, more preferably between 0.2 and 2 mm, and even more preferably between 0.3 and 1 mm. The thickness of said space denotes, in a circumferential plane, the shortest distance separating the two moulding elements. It is necessary for the moulding elements to be sufficiently far apart to prevent any risk of impact therebetween and sufficiently close together to minimize the thickness of the flash and thus to make it easier to cut off.

(15) According to a preferred embodiment, the mould comprises at least one sipe blade **40** that extends over two adjacent sectors **18** such that said sipe blade comprises a part in each of the two

sectors. In other words, said sipe blade comprises two parts **40a**, **40b**, said moulding elements **24** taking the form of each part of the sipe blade and, optionally, of the sector **18** to which a sipe blade part is joined. It is particularly appropriate to provide a space **26** and cutting means **38** on the particularly fragile edges of a “sectored” sipe blade. The sipe blade has a thickness of between 0.2 and 20 mm and, more preferably, between 0.5 and 5 mm.

(16) Preferably, the cutting means have the shape of a triangular or crenellated toothing. Such a shape is particularly easy to arrange on the edge **36** of a sipe blade **40**.

(17) The different moulding elements **24** and the cutting means **38** that are used in the composition of a mould **10** are made from alloys of steel, aluminium, polymer resin or some other rigid material suitable for such a use.

(18) The different moulding elements **24** and the cutting means **38** are obtained using a process chosen from the following group of processes: casting, pressing, laser cutting, laser, electrical discharge machining or consolidation by selective melting.

(19) “Selective consolidation by melting” is understood to be an additive manufacturing process intended to aggregate or agglomerate a feed of incoming work material progressively and selectively so as to obtain an outgoing work material. The incoming work material takes the form of and/or is used in the composition of a powder, of a wire or of a solution/bath. The incoming work material is generally introduced by depositing a powder on a support such as to form a layer. The support takes the form of a plate or a previously agglomerated layer. The agglomeration is generally achieved by solidification of the incoming work material, which is melted completely or partially (sintering) by localized or generalized input of energy, and then cooled. The input of energy is generally realized by a laser or an electron beam, although it can be realized by infrared radiation or induction. In the case of the laser and the electron beam, the localization of the input of energy is achieved by means for orienting the input of energy, such as optical or electromagnetic means, respectively. The process provides the incoming work material with a shape and mechanical characteristics that are predetermined. Said shape and said characteristics depend on the material of the incoming work material and on parameters of the process. Said shape is generally a one-piece solid, although it can be made up of a plurality of one-piece solids.

(20) Further variants and embodiments of the invention may be envisaged without departing from the scope of these claims.

Claims

1. A tire curing mold comprising: a plurality of circumferentially distributed sectors that are intended to mold a tread of a tire; and at least one sipe blade intended to mold a void in the tread, the sipe blade extending over two adjacent sectors such that the sipe blade comprises a part in each of the two sectors, the two parts of the sipe blade being separated, in a closed position of the mold, by a space that creates flash during curing of the tire, the two parts of the sipe blade comprising cutting means that are able to cut off the flash during opening of the mold.
2. The tire curing mold according to claim 1, wherein each of the parts of the sipe blade comprises an edge partially delimiting the space, the cutting means being arranged on the edge of at least one of the parts of the sipe blade such that the cutting means cut off the flash during withdrawal from the tire, each of the parts of the sipe blade being able to be withdrawn during the opening of the mold.
3. The tire curing mold according to claim 2, wherein the cutting means are arranged on the edge of each of the parts of the sipe blade, the cutting means of each of the parts of the sipe blade having complementary shapes.
4. The tire curing mold according to claim 2, wherein the cutting means are integral with the edge of at least one of the parts of the sipe blade.
5. The tire curing mold according to claim 1, wherein the cutting means have a recessed and/or

protruding shape.

6. The tire curing mold according to claim 1, wherein the space has a length of between 0.2 and 20 mm.

7. The tire curing mold according to claim 1, wherein the space has a thickness of between 0.1 and 3 mm.

8. The tire curing mold according to claim 1, wherein the sipe blade has a thickness of between 0.2 and 20 mm.

9. The tire curing mold according to claim 1, wherein the cutting means have the shape of a triangular or crenellated toothing.

10. The tire curing mold according to claim 1, wherein the cutting means are obtained by way of a process of consolidation by selective melting.

11. A method for manufacturing a tire comprising a step of curing the tire in the tire curing mold according to claim 1.
